



Habib University
CS 101 – Programming Fundamentals
Fall 2022
Python Programming in Interactive Mode
Practice Exercises

Question# 01.

Print following pattern using single print statement (Hint: Use \n inside a print statement):

```
PIZZA
PIZZ
PIZ
PI
P
```

Question# 02.

Let's try using Single and Double Quote in a single print statement:

```
print("Let's read Chapter 1 of our Course Textbook – Think Python.")
```

Question# 03.

Let's try printing String and Number together without space (Hint: Use str() function to convert 1 to string).

Print the following on screen: You are Number#1

Question# 04.

Try executing these in python's interactive mode. Pay attention to any error messages that you get and try to modify your command so that the error goes away.

- | | |
|---------------|------------|
| a) float("4") | j) 4//2.0 |
| b) int("4.5") | k) 4.0/2.0 |
| c) float(4) | l) 4.0/2 |
| d) int(4.67) | m) 4/2.0 |
| e) str(4.5) | n) 1+True |
| f) str(4) | o) 1+False |
| g) str(True) | p) 0+True |
| h) 4.0//2.0 | q) 16//6 |
| i) 4.0//2 | r) 18/6 |

Question# 05.

Read Chapter# 01 of your Course textbook ([Think Python](#)) and revise your lecture notes for greater insight into this week's content. Solve exercises given at the end of the chapter ([page 7-8](#)). The excerpt from the exercises (with few clarifications/examples added) are given here:

Exercise 1.1.

1. In a print statement, what happens if you leave out one of the parentheses, or both?

```
print("Hello World!"
```

```
print "Hello World!")
```

```
print "Hello World!"
```

2. If you are trying to print a string, what happens if you leave out one of the quotation marks, or both?

```
print(Hello World)
```

```
print('Hello World!)
```

```
print('Hello World!")
```

```
print("Hello World!)
```

```
print(Hello World! ')
```

```
print(Hello World!")
```

```
print("Hello World!")
```

3. You can use a minus sign to make a negative number like -2. What happens if you put a plus sign before a number? What about 2++2?

4. In math notation, leading zeros are ok, as in 09. What happens if you try this in Python? What about 011?

5. What happens if you have two values with no operator between them? The two values are separated by a space.

```
2 2
```

```
"Hello" "World"
```

```
4.5 2
```

```
"Hello" 2
```

Exercise 1.2. Start the Python interpreter and use it as a calculator.

NOTE: you should calculate the value (using arithmetic expression) in python's interactive mode and not "hard-code" the result.

1. How many seconds are there in 42 minutes 42 seconds?

2. How many miles are there in 10 kilometers? Hint: there are 1.61 kilometers in a mile.

3. If you run a 10 kilometer race in 42 minutes 42 seconds, what is your average pace (time per mile in minutes and seconds)? What is your average speed in miles per hour?